

**SEMINAR SUMMARY:
FEBRUARY 24, 2026 - LAURA FIX
PATIENT-SPECIFIC MODELING OF CARDIOVASCULAR AND
RESPIRATORY DYNAMICS**

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1. MODELING OF CARDIOPULMONARY DYNAMICS

The utilization of a mathematical model in regard to modeling the dynamics of blood-flow through the body is informative and understood to be fairly predictive of the actual dynamics of the physiological system at hand. A standard in the field of fluid dynamics for modeling flow through a network of pipes, pumps, check valves, capacitor tanks. Ohms law in fluid dynamics follows as

$$P = Q \cdot R$$

where P is pressure head at a spot along the pipe, Q is the flow rate, and R is the resistance of the pipe. Now this is a description of the model that we were presented and is an ODE model. Generally the system can be modeled with ODEs or PDEs.

We were given two projects that were being actively worked on: prediction of dynamic cerebrovascular resistance and patient-specific modeling of respiratory mechanics in preterm infants (which seems to have new material to work on).